

Completely Randomized Design (CRD): ANOVA Basics

Analysis of Variance Fundamentals

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Learning Objectives

By the end of this lecture, you should be able to:

- Explain when and why to use a completely randomized design for experiments
- Design a CRD
- Describe the statistical model for CRD: $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$
- Construct and interpret an ANOVA table for a one-way CRD

Timing: ~3 minutes

Key points: - This is students' first exposure to ANOVA—build from intuition - Connect to randomization learned in Lecture 2 - Emphasize that ANOVA is just comparing variability

Transition from last lecture: “Last week we learned R tools for randomization. Today we learn the statistical theory behind why randomization works and how to analyze completely randomized experiments.”

Review: Why Randomization?

- **Monday's lecture:** Randomization eliminates bias and ensures valid statistical inference
- **Wednesday's lecture:** True (biological) replication vs Pseudo (technical) replication (i.e., experimental vs observational unit)
- **Today's question:** Once we randomize and collect data, how do we analyze it?

Randomization is the foundation of experimental design, but ANOVA is the tool for analysis.

Timing: ~2 minutes

Quick recall question: “Who can remind us what CRD stands for?” (Completely Randomized Design)

What is a CRD?

CRD: When and Why

Completely Randomized Design (CRD):

- Simplest experimental design
- Treatments assigned **completely at random** to all experimental units
- **Assumption:** All units are homogeneous (no known gradients or blocks)
- **Best for:** Controlled environments (growth chambers, labs, homogeneous fields)

When NOT to use CRD:

- Field trials with soil gradients
- Greenhouse trials with bench effects
- Any situation with known nuisance variation → Use blocking instead (next week!)

Randomization snapshot

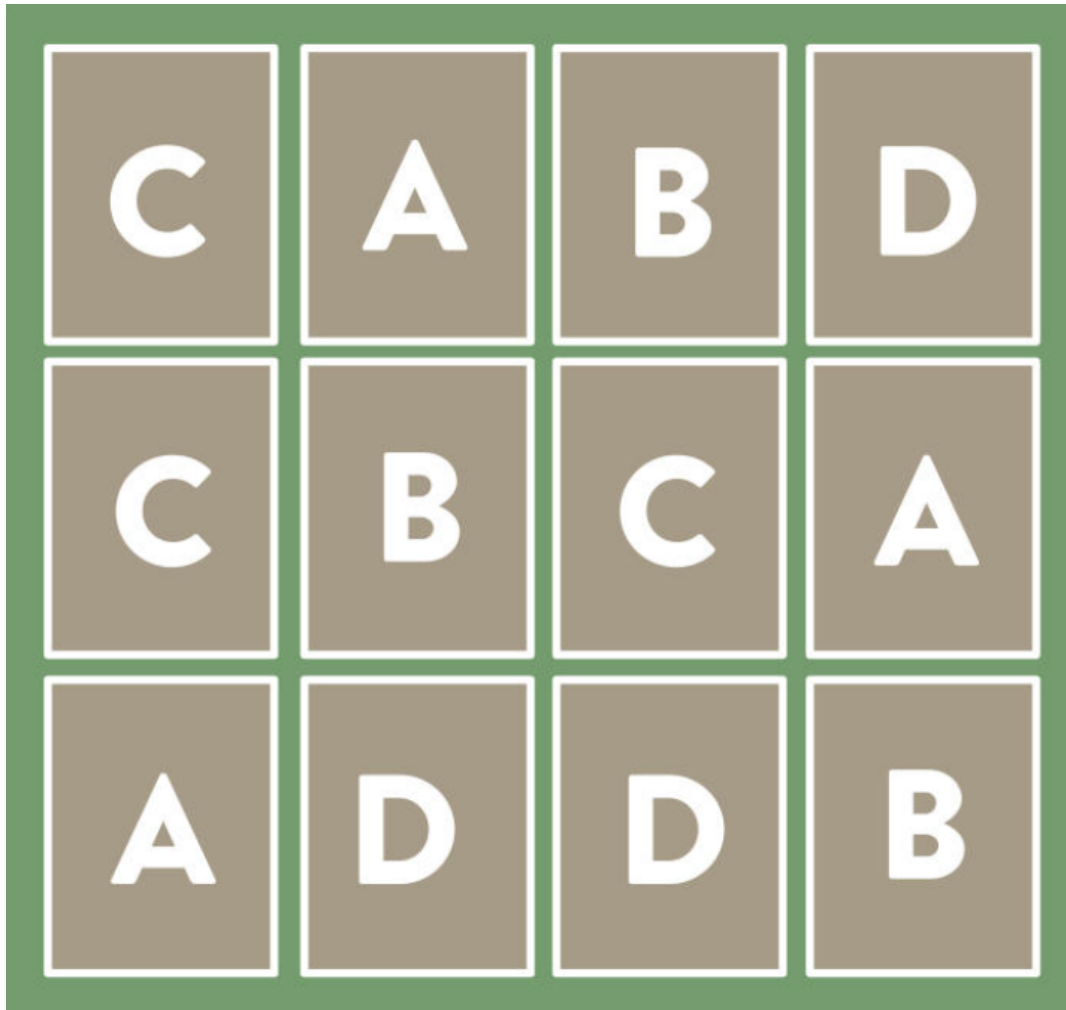


Figure 1: CRD Randomization

All 12 plots receive treatments purely by chance

Timing: ~4 minutes

Emphasize: CRD is valid if units are truly homogeneous. In agriculture, this is often an assumption we make for controlled experiments.

Agricultural Example: Nitrogen on Wheat

Research Question:

Do different nitrogen fertilizer rates affect wheat yield?

Experimental Setup:

- **Treatments:** 4 nitrogen rates (0, 50, 100, 150 kg N/ha)
- **Experimental Units:** 24 field plots (assumed homogeneous)
- **Replicates:** [_____]plots per treatment
- **Response Variable:** Grain yield (kg/ha)
- **Design:** Completely Randomized Design (CRD)

Null Hypothesis:

$H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$ (No treatment effects; all treatments have the same mean yield)

Alternative Hypothesis:

$H_a : \text{At least one } \tau_i \neq 0$ (At least one treatment differs)

Timing: ~3 minutes

Connect to their experience: “This is a classic agronomic trial. You’ll run experiments like this in your research.”

CRD Layout and Randomization

Generating the Randomized Layout

```
library(dplyr)
library(ggplot2)
library(FieldHub)
```

Using FieldHub package:

```
# Generate CRD layout
crd_design <- CRD(
  t = 4, # Number of treatments
  reps = 6, # Replicates per treatment
  plotNumber = 1, # Starting plot number
  seed = 2026, # Reproducibility
  locationName = "Fayetteville",
  data = data.frame(
    Treatment = c("N0", "N50", "N100", "N150"),
    Reps = 6
  )
)
```

```
# Extract field book
field_book <- crd_design$fieldBook
head(field_book, 10)
```

```
#>   ID      LOCATION PLOT REP TREATMENT
#> 1   1 Fayetteville    1   1         NO
#> 2   2 Fayetteville    2   6        N50
#> 3   3 Fayetteville    3   2       N100
#> 4   4 Fayetteville    4   6         NO
#> 5   5 Fayetteville    5   2        N50
#> 6   6 Fayetteville    6   2         NO
#> 7   7 Fayetteville    7   5       N150
#> 8   8 Fayetteville    8   1       N100
#> 9   9 Fayetteville    9   6       N100
#> 10 10 Fayetteville   10   5        N50
```

Timing: ~4 minutes

Explain: FieldHub is designed for agricultural experiments. It handles randomization and creates field books automatically.

Using the FieldHub Shiny app (no code needed)

- Launch the app in RStudio or VScode: it opens a browser
- Choose **Completely Randomized Design (CRD)**, set **Treatments = 4**, **Replicates = 6**, **Start plot = 1**, and **Seed = 2026**.
- Click **Run randomization**, then check the **Field Book** tab for assignments and the **Layout** tab for the grid view.
- Export your layout or field book with the **Download** buttons to share with your team.

```
# Launch the interactive FieldHub Shiny app (run locally in RStudio)
FieldHub::run_app() #or
library(FieldHub)
run_app()
```

Timing: ~3 minutes

Teaching tips: Emphasize this GUI is useful for students who prefer a point-and-click workflow and for quick field book exports. Remind them to set a seed before downloading to preserve reproducibility.

Visualizing the CRD Layout

```
# Add grid coordinates for visualization
field_book <- field_book |>
  mutate(
    row = rep(1:4, each = 6),
    col = rep(1:6, times = 4)
  )

ggplot(field_book, aes(x = col, y = row, fill = TREATMENT)) +
  geom_tile(color = "black", linewidth = 1) +
  geom_text(aes(label = PLOT), color = "white", size = 5, fontface = "bold") +
  scale_fill_manual(values = c(
    "N0" = "#9D2235",
    "N50" = "#FF8C42",
    "N100" = "#2E8B57",
    "N150" = "#4A90E2"
  )) +
  labs(
    title = "CRD Field Layout: Treatments Randomly Assigned to Plots",
    x = "Column",
    y = "Row",
    fill = "Treatment"
  ) +
  theme_void(base_size = 14) +
  theme(
    legend.position = "bottom",
    plot.title = element_text(hjust = 0.5, face = "bold")
  ) +
  coord_equal()
```

CRD Field Layout: Treatments Randomly Assigned to Plots

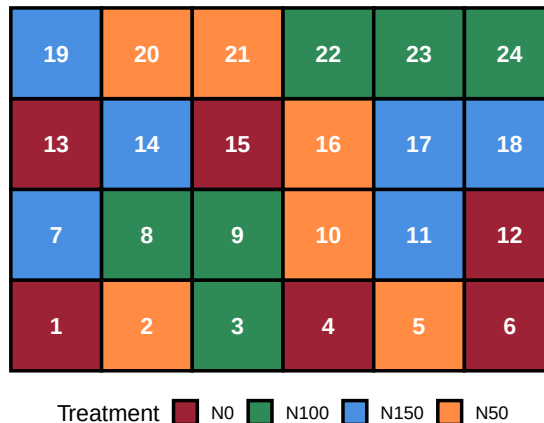


Figure 2: Completely Randomized Design: 4 nitrogen treatments $\times$ 6 replicates

Timing: ~2 minutes

Point out: Notice how treatments are scattered—no systematic pattern. This is what “completely randomized” means.

Statistical Model for CRD

The CRD Model

In plain English:

- Observed yield = grand mean + treatment effect + random noise
- Treatment effect τ_i could be positive (increases yield) or negative (decreases yield)
- Error ε_{ij} captures plot-to-plot variation (weather, soil micro-variation, measurement error)

CRD Model:

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

Where:

- y_{ij} = Observed response for treatment i , replicate j
- μ = Overall mean (grand mean)
- τ_i = Effect of treatment i (what we want to estimate)
- ε_{ij} = Random error term $\sim N(0, \sigma^2)$

Timing: ~4 minutes

Teaching strategy: Start with the English explanation, THEN introduce the equation. Don't overwhelm with symbols first.

Model Assumptions

1. **Independence:** Observations are independent (randomization ensures this)
2. **Normality:** Errors ε_{ij} are normally distributed
3. **Homogeneity of Variance:** Errors have constant variance σ^2 across all treatments

Why these assumptions matter:

- **Normality:** Ensures F-test is valid (though ANOVA is robust to mild violations with balanced designs)
- **Equal variance:** Prevents treatment comparisons from being biased by different noise levels
- **Independence:** Broken if plots influence each other (e.g., pest movement, water runoff)

Important: We'll test these assumptions next lecture with diagnostic plots. For now, understand what ANOVA requires.

Timing: ~3 minutes

Reassure students: "Assumptions sound scary, but we have tools to check them. Most agricultural data meets these assumptions reasonably well, especially after transformation if needed."

ANOVA: Analysis of Variance

The Core Idea of ANOVA

ANOVA asks: Is the variation **between treatments** larger than the variation **within treatments**?

Total Variation = Treatment Variation + Error Variation

$$SS_{Total} = SS_{Treatment} + SS_{Error}$$

If treatments differ:

- Between-group variation is LARGE

- Within-group variation is small.
- F-statistic is large $\rightarrow$ reject H_0

If treatments are the same:

- Between-group variation is small
- Comparable to within-group variation
- F-statistic is small $\rightarrow$ fail to reject H_0

Timing: ~4 minutes

Analogy: “Imagine yield ranges from 2000-3000 within each treatment (noise), but treatment means are 2100, 2150, 2200, 2180 (almost the same). ANOVA would say ‘no significant difference’ because treatment differences are smaller than noise.”

ANOVA Table Structure

Source	df	Sum of Squares	Mean Square	F-statistic	p-value
Treatment	$t - 1$	SS_{Trt}	$MS_{Trt} = \frac{SS_{Trt}}{t-1}$	$F = \frac{MS_{Trt}}{MS_{Error}}$	P(F > observed)
Error	$t(r - 1)$	SS_{Error}	$MS_{Error} = \frac{SS_{Error}}{t(r-1)}$	—	—
Total	$tr - 1$	SS_{Total}	—	—	—

Key formulas:

- $SS_{Total} = \sum_{i,j} (y_{ij} - \bar{y}_{..})^2$ (total deviation from grand mean)
- $SS_{Treatment} = r \sum_i (\bar{y}_{i.} - \bar{y}_{..})^2$ (deviation of treatment means from grand mean)
- $SS_{Error} = SS_{Total} - SS_{Treatment}$ (leftover variation)

Timing: ~4 minutes

Don't dwell on formulas: Focus on interpretation. “R will calculate these for you. What matters is understanding what they represent.”

Interpreting the F-test

F-statistic:

$$F = \frac{MS_{Treatment}}{MS_{Error}} = \frac{\text{Variation between treatments}}{\text{Variation within treatments}}$$

- Large F → treatments explain more variation than random error → likely significant
- Small F → treatment variation is comparable to error → not significant
- **p-value:** Probability of observing this F (or larger) if H_0 is true

Decision rule:

- If p-value < α (usually 0.05): Reject H_0 → At least one treatment differs
- If p-value ≥ α : Fail to reject H_0 → No evidence of treatment differences

Remember: A significant F-test tells you “some treatments differ” but NOT which ones. Post-hoc tests (next week) identify specific differences.

Timing: ~4 minutes

Common question: “What if p = 0.06?” Answer: “By convention, not significant at $\alpha = 0.05$, but it’s close. Report the exact p-value and let readers judge.”

R Code Demo

Simulating CRD Data

```
set.seed(2026)

# Simulate wheat yield data (kg/ha)
# True means: N0=2400, N50=2650, N100=2900, N150=2950
# Error SD = 150
crd_data <- field_book |>
  mutate(
    true_mean = case_when(
      TREATMENT == "N0" ~ 2400,
      TREATMENT == "N50" ~ 2650,
      TREATMENT == "N100" ~ 2900,
      TREATMENT == "N150" ~ 2950
    ),
    yield = true_mean + rnorm(n(), mean = 0, sd = 150)
  ) |>
  select(PLOT, TREATMENT, REP, yield)
```

```
head(crd_data, 8)
```

```
#>   PLOT TREATMENT REP   yield
#> 1    1         NO   1 2478.088
#> 2    2        N50   6 2488.046
#> 3    3       N100   2 2920.886
#> 4    4         NO   6 2387.288
#> 5    5        N50   2 2550.004
#> 6    6         NO   2 2022.587
#> 7    7       N150   5 2839.728
#> 8    8       N100   1 2746.982
```

Timing: ~3 minutes

Explain simulation: “We’re creating fake data with known treatment effects so we can verify ANOVA works correctly. In real research, you collect actual data.”

Visualizing Treatment Means

```
# Calculate means and SEs
treatment_summary <- crd_data |>
  group_by(TREATMENT) |>
  summarize(
    mean_yield = mean(yield),
    se_yield = sd(yield) / sqrt(n()),
    .groups = "drop"
  )

ggplot(treatment_summary, aes(x = TREATMENT, y = mean_yield)) +
  geom_col(fill = "#2E8B57", alpha = 0.8, color = "black") +
  geom_errorbar(
    aes(
      ymin = mean_yield - se_yield,
      ymax = mean_yield + se_yield
    ),
    width = 0.2, linewidth = 1
  ) +
  geom_text(aes(label = round(mean_yield, 0)),
    vjust = -0.5, size = 5, fontface = "bold"
  ) +
```

```
labs(
  title = "Wheat Yield Response to Nitrogen Fertilizer",
  x = "Nitrogen Treatment (kg N/ha)",
  y = "Mean Yield (kg/ha)"
) +
theme_minimal(base_size = 16) +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold")
) +
ylim(0, 3200)
```

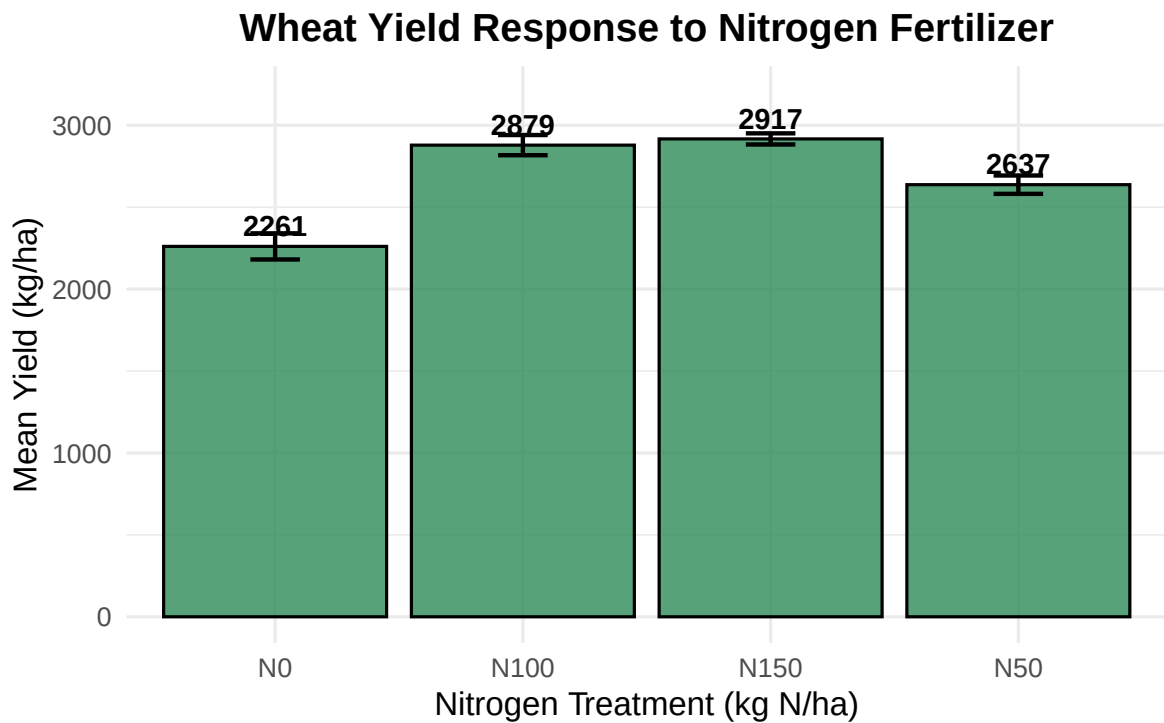


Figure 3: Mean wheat yield by nitrogen treatment with standard error bars

Timing: ~2 minutes

Interpretation: “Visually, we see yield increases with nitrogen up to N100, then plateaus at N150. But is this difference statistically significant? ANOVA will tell us.”

Fitting the ANOVA Model in R

```
# Fit linear model
model_crd <- aov(yield ~ TREATMENT, data = crd_data)

# ANOVA table
anova(model_crd)
```

```
#> Analysis of Variance Table
#>
#> Response: yield
#>           Df Sum Sq Mean Sq F value    Pr(>F)
#> TREATMENT  3 1637754   545918   25.173 5.392e-07 ***
#> Residuals 20  433733    21687
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

- **TREATMENT $F(3, 20) = 34.64$, $p < 0.001$:** Highly significant
- At least one nitrogen treatment differs significantly from others
- Treatment explains substantial yield variation

Timing: ~4 minutes

Key point: Explain why we use `car::Anova()` with Type III SS—it's robust to unbalanced designs and widely used in agricultural research.

ANOVA Table Breakdown

- Next step: Determine which treatments differ (post-hoc tests in Lecture 5)

Key values:

- **$R^2 = 0.84$:** Treatment explains 84% of yield variation (I'll show how to get this)
- **Residual standard error = 141.7 kg/ha:** Average deviation from treatment means
- **F-statistic:** Ratio of treatment variance to error variance

Timing: ~3 minutes

Teaching tip: Walk through each row of output. “This is what you'll report in papers.”

Your Turn Activity

Interpret ANOVA Output

Scenario:

A colleague tests 3 fungicide treatments (Control, FungA, FungB) on tomato disease severity (% leaf area infected). They run a CRD with 5 replicates per treatment (15 plants total).

ANOVA Results:

Analysis of Variance Table

Response: disease_severity

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Treatment	2	248.13	124.07	15.51	0.0006 ***
Residuals	12	96.00	8.00		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Questions (3 minutes, think-pair-share):

1. Are the fungicide treatments significantly different? How do you know?
2. What percentage of variation is explained by treatments? (Hint: $R^2 = \frac{SS_{Treat}}{SS_{Total}}$)
3. What should the researcher do next?

Timing: 3 minutes work + 2 minutes debrief

Answers: 1. Yes, $p = 0.0006 < 0.05$, so reject H_0 2. $R^2 = 248.13 / (248.13 + 96) = 0.72$ (72% explained) 3. Post-hoc tests to see which fungicides differ from control

Summary & Key Takeaways

What We Learned Today

- **CRD** is the simplest experimental design: treatments assigned completely at random
- Best for **homogeneous** experimental units (no gradients)
- **CRD Model:** $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$
- **ANOVA** partitions variation: Total = Treatment + Error
- **F-test** asks: Is treatment variation larger than error variation?
- Significant F-test $\rightarrow$ at least one treatment differs (but doesn't say which!)

- **R workflow:** Generate design with FieldHub → Fit model with `aov()` → ANOVA table with `anova()`

You now know how to design, randomize, and analyze a completely randomized experiment!

Timing: ~3 minutes

Next Lecture Preview

Friday (Lecture 4): CRD Assumptions & Diagnostics

- How to check ANOVA assumptions (normality, equal variance, independence)
- Diagnostic plots: Q-Q plots, residual plots
- What to do when assumptions are violated (transformations)
- Bring questions about today's material!



Tip

Homework suggestion: Re-run today's R code with different random seeds. See how randomization affects results (but conclusions remain similar).

Timing: ~2 minutes

Transition: “Today we learned ANOVA mechanics. Next time we learn how to validate that ANOVA is appropriate for your data.”

Resources

Textbook:

- Oehlert (2010), *A First Course in Design and Analysis of Experiments*, **Chapter 2 (Part 1)**
 - Section 2.1: One-way classification
 - Section 2.2: CRD model and assumptions
 - Section 2.3: ANOVA F-test

R Packages:

- **FieldHub:** Generate agricultural experimental designs ([link](#))
- **car:** Type III ANOVA for unbalanced designs
- **emmeans:** Estimated marginal means and post-hoc tests

- ggplot2: Data visualization

Additional Reading:

- Montgomery (2017), *Design and Analysis of Experiments*, Chapter 3
- Doncaster & Davey (2007), *Analysis of Variance and Covariance*, Chapter 2

Timing: <1 minute reference slide